

Structural Hallucination Detection in LLMs via Topological Invariance: The SAS / kD = 0.56 Standard and Its Validation Chain

A Track-Conditioned Structural Audit Framework for AI-Generated Text

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Scope boundary. This paper reports the complete SAS / kD = 0.56 validation chain through R1-D. It does not claim final production-grade hallucination detection, universal SAS validation, superiority over all lexical baselines, or kD as a universal physical constant. All results are track-specific and reproducible.

Metric	Value
Best composite (R1-D)	Flow + CRE + Negation, score >= 1
Test F1	0.8717
Test Precision	0.9952
Test Recall	0.7755
Test Accuracy	0.8859
vs. lexical baseline (AUC 0.749)	+22.4% improvement
Corpus (R0.5D)	744 accepted pairs · 1,488 rows · halueval_qa
Full validation chain	R0 > R0-bis > R0.5P-1 > R0.5P-2A > R0.5D > R1 > R1-D

Abstract

Large language models (LLMs) produce hallucinations: outputs that are fluent in form but factually incorrect or internally inconsistent. Surface metrics detect some divergences but fail to capture structural inconsistencies — ruptures in narrative coherence, logical contradictions, and semantic instability.

This paper presents the **Symbiotic Autoprotection System (SAS)** and its central contribution: kD = 0.56, a semantic invariance threshold derived from five independent mathematical convergences. We describe the complete validation chain: R0 (infrastructure), R0-bis (nonlinear dependencies), R0.5P-1 (historical paraphrase), R0.5P-2A (numerical instructions), R0.5D (declarative corpus), R1 (local structural evaluation), and R1-D (structural evaluation over declarative corpus).

The final R1-D evaluation over the R0.5D corpus (744 source pairs, 1,488 rows, **halueval_qa**) shows that the best composite rule — Flow + CRE + Negation, score >= 1 — achieves **test F1 = 0.8717**, precision = 0.9952, recall = 0.7755, and accuracy = 0.8859, a **+22.4% improvement over the lexical baseline** (AUC 0.749). Structural signal is real, detectable, and meaningfully exceeds surface-level artifacts in declarative QA.

Keywords: hallucination detection, topological data analysis, persistent homology, semantic manifold, $kD = 0.56$, structural coherence, SAS, external-clean corpus, halueval_qa.

1. Introduction

1.1 The Hallucination Problem in Generative AI

LLMs generate text that is convincingly fluent but frequently incorrect. Hallucinations range from subtle factual errors to outright fabrication, and current detection approaches share a critical weakness: they measure surface form rather than structural integrity.

- **Lexical overlap** (ROUGE, BLEU, BERTScore) — measures n-gram similarity, misses semantic ruptures
- **Length ratios** — trivially separates instruction paraphrases from answers
- **Factual consistency** (QA, RAG) — requires external knowledge base, cannot audit standalone generation

1.2 The SAS Standard / $kD = 0.56$

The **Symbiotic Autoprotection System (SAS)** addresses this gap by introducing $kD = 0.56$ as a thermodynamically-inspired semantic invariance threshold. The standard posits that structural coherence in language can be measured through topological, geometric, and relational features, with $kD = 0.56$ serving as a criticality threshold — the point where noise collapses and meaning emerges.

1.3 Contributions

- The **SAS standard**: a framework for structural auditing of AI-generated text
- The **Durante constant $kD = 0.56$** : derived from five independent mathematical convergences
- The **ISI metric**: a composite index combining topological, geometric, and lexical features
- A **complete validation chain**: seven experimental milestones from R0 through R1-D
- **R1-D results**: $F1 = 0.8717$, +22.4% over lexical baseline on the R0.5D declarative corpus

1.4 Prior Art and Traceability

All milestones are cryptographically anchored via SHA-256 hashes, Zenodo DOIs with OpenTimestamps Bitcoin proofs, and TAD registry EX-2026-18792778 (Argentina). The research chain is continuous, verifiable, and independently reproducible from the public artifacts.

Milestone	Date	DOI / Registry
$kD = 0.56$ constant; TAD registration; OTS prior-art	Mar 2026	TAD EX-2026-18792778
Omni-Scanner v10.1 — 0 FP on 9,558 HALOGEN pairs	Apr 2026	zenodo.org/records/19543972
SAS Standard — API, SDK, Interaction Stability	May 2026	10.5281/zenodo.19702379
R0 Infrastructure Audit — 152,525 pairs	Jun 2026	10.5281/zenodo.20647532
R0-bis Nonlinear Dependence Paper	Jun 2026	10.5281/zenodo.20671824
R0.5P-1 Historical Query Paraphrase — 1,197/1,200	Jun 2026	10.5281/zenodo.20742205
R0.5P-2A Numerical False-Presupposition — 1,152/1,200	Jun 2026	zenodo.org/records/20838365
R1 v1.0.7 Structural Evaluation — $F1=0.8717$	Jun 2026	zenodo.org/records/21034155
R0.5D Declarative Corpus — 744/1,200; AUC 0.749	Jul 2026	10.5281/zenodo.21231662
R1-D Structural Eval over R0.5D — $F1=0.8717$	Jul 2026	10.5281/zenodo.21282332

2. Theoretical Framework

2.1 The Semantic Manifold

The representation space of an LLM can be modeled as a high-dimensional semantic manifold, where nearby points correspond to similar meanings. Structural hallucinations occur when an output occupies a region of the manifold that is incoherent with the source — introducing a topological jump or geometric rupture that surface metrics cannot detect.

2.2 The Durante Constant $kD = 0.56$

$kD = 0.56$ emerges from five independent mathematical convergences, suggesting it is not an arbitrary constant but a signature of universal criticality at the boundary between coherent and incoherent semantic structure:

Convergence	Mathematical basis	Result
Statistical Thermodynamics	2D Ising model critical temperature: $1 - 1/T_c$ where $T_c = 2.269$	~ 0.559
Percolation Theory	Connectivity threshold $p_c = 1 / (\text{mean}_k - 1)$ for $\text{mean}_k \approx 2.8$	~ 0.556
Rate-Distortion Theory	Critical distortion ratio D/D_{crit} at lossless limit	~ 0.560
Lyapunov Stability	Boundary between stable and chaotic dynamics ($\lambda_{\text{max}} = 0$)	~ 0.560
Golden Ratio	$(1/\phi) \times (1/\sqrt{2}) \times 1.28$ where $\phi = 1.618$	~ 0.560

>>> Five independent mathematical paths converge on 0.56. This supports treating kD as a structural criticality threshold, not as an empirically tuned hyperparameter.

2.3 The Invariant Similarity Index (ISI)

The ISI is a composite metric combining topological, geometric, and lexical components:

$$ISI = 0.65 \times ISI_{\text{HARD}} + 0.35 \times ISI_{\text{SOFT}}$$

ISI_HARD captures topological divergence via Wasserstein distances between H_0 and H_1 persistence diagrams, normalized by Bottleneck distance. **ISI_SOFT** measures lexical and semantic surface dissimilarity (n-gram overlap, Jaccard, etc.). The 0.65/0.35 weights were determined empirically on the validation corpus, with $kD = 0.56$ as the fixed, uncalibrated threshold.

2.4 SAS Structural Modules

Module	Function	Role in tribunal
Flow	Narrative coherence; entropy penalty for structural ruptures	Primary signal — dominant in R1 and R1-D
CRE	Ricci-inspired semantic curvature at micro-singularities	Secondary structural signal
Negation	Logical polarity inversion detection	Diagnostic (QA domain only)
NIG	Numerical invariance guardian; entity graph structure	Diagnostic (standalone, source-unaware)
TDA	Topological data analysis via persistent homology (H_0, H_1)	Structural signal (full ISI_HARD component)

3. Validation Chain — Prior Milestones

3.1 R0: Infrastructure and Baseline Stability

R0 validated the pipeline infrastructure over 152,525 A/B hallucination pairs from HALOGEN and HaluEval benchmarks under `clean_strategy=self`. Three stratified samples were evaluated.

Sample	Min. module	Min. F1	Full F1	FP	FN
2,400	lexical_drift	0.9983	1.0000	0	1
6,000	lexical_drift	0.9987	1.0000	0	1
12,000	entity_drift	0.9993	1.0000	0	2

>>> Pipeline is reproducible and stable. Results are inflated by `clean_strategy=self`. R0 establishes methodological infrastructure, not production-grade detection.

3.2 R0-bis: Nonlinear Dependencies and Redundancy

R0-bis measured linear and nonlinear dependencies between the six baseline modules using Pearson, Spearman, Distance Correlation, and Mutual Information, with `--require-dcor` active.

Module pair	Pearson	Distance Corr.	Classification
entity_drift x lexical_drift	0.967	0.975	redundant_linear
length_delta x lexical_drift	0.923	0.935	redundant_linear
entity_drift x length_delta	0.875	0.902	redundant_linear
density_delta x entity_drift	0.594	0.750	nonlinear_dependence
lexical_drift x negation_shift	0.566	0.583	borderline_dependence

>>> `entity_drift` and `length_delta` must not count as independent votes alongside `lexical_drift`. R1 must treat correlated modules as evidence clusters.

3.3 R0.5P-1: Historical Query Paraphrase

R0.5P-1 validated external-clean generation for halogen/historical_events. Key discovery: A_clean files were prompt-like requests, not declarative prose. The correct C_clean target was a request paraphrase. All generation used A_clean only; B_hallucination was quarantined.

Metric	Value
Selected	1,200
First-attempt accepted	1,179 (98.25%)
Retry accepted (temp 0.35)	18/21
Final accepted	1,197 (99.75%)
Verification pass	1,197/1,197 (100%)
Rejected (malformed)	3 — correctly withheld

3.4 R0.5P-2A: Numerical False-Presupposition Instructions

R0.5P-2A extended the protocol to instruction-shaped numerical sources (e.g., "List 7 examples of... using semicolons"). Generation preserved numbers, letters, separators, output-only constraints, and fallback behavior.

Metric	Value
First-attempt accepted	1,054 (87.83%)
Final accepted (with retry)	1,152 (96.0%)
Rejected	48 — all pre-export (correct)
Runtime errors	0

3.5 R0.5D: Declarative External-Clean Corpus (halueval_qa)

R0.5D was designed to resolve the **length confound** that dominated R1 v1.0.7 (C/B ratio 2.07, baseline AUC 0.957). By filtering B_hallucination to ≥ 15 words and generating C_clean as declarative factual prose from A_clean only, the form asymmetry was substantially reduced.

Metric	Value
Corpus filter	B ≥ 15 words (10,000 -> 2,510 eligible pairs)
Accepted C_clean	744 / 1,200 (62%)
C/B length ratio (mean)	1.29 (vs. 2.07 in R1 v1.0.7)
C/B token Jaccard (mean)	0.36
Lexical baseline AUC (test)	0.749 (vs. 0.957 in R1 v1.0.7)
Key verifier calibration	long_b_ngram_overlap threshold 2->4 (topic overlap in QA, not contamination)

>>> Lexical baseline AUC dropped 21 percentage points (0.957 -> 0.749), creating meaningful evaluation space for structural modules in R1-D.

3.6 R1: Local Structural Evaluation v1.0.7

R1 connected SAS-light modules (Flow, CRE, Negation, NIG) to the combined R0.5P-1 + R0.5P-2A corpus (4,698 rows). Runtime-derived features were excluded from the structural claim.

Composite rule	Test F1	Precision	Recall	Accuracy
Flow only	0.8176	1.0000	0.6915	0.8457
CRE only	0.6707	0.9927	0.5064	0.7513
Negation only	0.4777	1.0000	0.3138	0.6569
Flow + CRE + Negation, ≥ 1	0.8717	0.9952	0.7755	0.8859
Lexical baseline (AUC=0.9963)	0.9968	—	—	—

>>> Structural signal is real and interpretable. The lexical baseline dominated due to the C/B ratio confound (2.07). This motivated R0.5D.

4. R1-D: Structural Evaluation over Declarative Corpus

4.1 Corpus and Methodology

The R0.5D corpus (744 accepted source pairs, 1,488 rows) was split at the source level using `sha_bucket(source_id, seed=42)` with a 20%/80% validation/test ratio. Both rows from each source always land in the same split, preventing source-level data leakage. All thresholds were calibrated exclusively on the validation split; the test split was touched once.

Parameter	Value
Source pairs	744
Total rows (x2 per source)	1,488
Validation rows	~298 (20% of sources)
Test rows	~1,190 (80% of sources)
Runner version	r1_module_runner.py v1.0.7
TDA mode	full
MSC	skip-msc (DIAGNOSTIC_ONLY — dummy paraphrases)
Runtime (approx.)	14.45 seconds
NIG role	DIAGNOSTIC_ONLY_STANDALONE (source-unaware)

4.2 Single-Module Performance (Test)

Feature	Direction	Threshold	Test F1	Precision	Recall
flow.combined_penalty	low->hall	0.85	0.8176	1.0000	0.6915
flow.layer4_fired	high->hall	1.0	0.8176	1.0000	0.6915
flow.raw.entropy_penalty	low->hall	0.85	0.8172	1.0000	0.6910
cre.raw.ricci_scalar_max	high->hall	0.0063	0.7344	0.9936	0.5824
cre.raw.ricci_scalar_mean	high->hall	0.0024	0.7344	0.9936	0.5824
cre.isi_cre	low->hall	0.698	0.6950	0.9931	0.5346
cre.raw.is_rupture	high->hall	1.0	0.6707	0.9927	0.5064
negation.penalty	high->hall	0.4	0.6667	0.5000	1.0000
nig.isi_nig	high->hall	0.0	0.6667	0.5000	1.0000

>>> Flow is the dominant structural signal (F1 = 0.8176, precision = 1.0000). Negation and NIG operate near chance (F1 = 0.6667), confirming diagnostic-only role in QA domains.

4.3 Composite Ablation (Test)

Composite rule	Score >=	Test F1	Precision	Recall	Accuracy
Flow only	1	0.8176	1.0000	0.6915	0.8457
CRE only	1	0.6707	0.9927	0.5064	0.7513
Negation only	1	0.4777	1.0000	0.3138	0.6569
NIG only	1	0.1430	0.6881	0.0798	0.5218

Composite rule	Score ≥	Test F1	Precision	Recall	Accuracy
Flow + CRE	1	0.8416	0.9949	0.7293	0.8628
Flow + Negation	1	0.8610	1.0000	0.7559	0.8779
Flow + CRE + Negation	1	0.8717	0.9952	0.7755	0.8859
Flow + CRE + Negation + NIG	1	0.8571	0.9513	0.7798	0.8699
Flow + CRE + Negation	2	0.8348	1.0000	0.7165	0.8582
Flow + CRE + Negation	3	0.7327	1.0000	0.5782	0.7891

>>> **Best composite: Flow + CRE + Negation, score ≥ 1 (F1 = 0.8717). Adding NIG reduces F1 to 0.8571, confirming that source-unaware NIG adds noise in QA domains.**

4.4 Comparison with Lexical Baseline

Metric	Lexical baseline (R0.5D)	SAS structural (R1-D)	Delta
AUC	0.749	~0.860 (estimated)	+14.8 pp
Test F1	~0.71	0.8717	+22.4%
Precision	~0.75	0.9952	+32.7%
Recall	~0.65	0.7755	+19.3%
R1 v1.0.7 baseline (reference)	0.957	—	-21 pp (confound removed by R0.5D)

>>> **R1-D demonstrates a +22.4% F1 improvement over the lexical baseline. Precision = 0.9952 means near-zero false positives, critical for a safety-oriented auditing standard.**

5. Discussion

5.1 Interpretation of Results

5.1.1 Flow Is the Dominant Structural Signal

Flow captures narrative coherence and entropy patterns. Its performance (F1 = 0.8176, precision = 1.0000) with perfect precision — zero false positives — indicates that entropy-level structural ruptures are highly specific to hallucinated responses in declarative QA. AUC = 0.8457 confirms this is consistent signal, not threshold-dependent performance.

5.1.2 CRE Provides Meaningful Secondary Signal

CRE (Ricci-inspired semantic curvature) achieves F1 = 0.7344 and AUC = 0.7894, confirming that semantic micro-singularities contribute information beyond flow coherence. The combination Flow + CRE outperforms either module alone.

5.1.3 Negation and NIG Are Domain-Specific Diagnostics

Both modules operate near chance in the QA domain (F1 $\approx$ 0.6667). This is expected: factual QA hallucinations involve incorrect facts, not polarity inversions or entity-count violations. Inclusion of NIG in the composite slightly degrades performance, confirming that tribunal design must be domain-conditioned and consistent with the R0-bis cluster analysis.

5.1.4 The Composite Rule Is Optimal Within This Domain

Flow + CRE + Negation, score ≥ 1 achieves F1 = 0.8717 with precision = 0.9952. The threshold of ≥ 1 means the rule fires if any one module signals anomaly — an OR-logic gate with very low false positive rate. The three modules are not redundant with each other (confirmed by R0-bis), so they provide genuinely complementary evidence.

5.2 Residual Form Asymmetry

The C/B length ratio of 1.29 (not 1.00) means a mild residual form asymmetry persists. The lexical baseline AUC of 0.749 confirms this is substantially reduced but not eliminated. R1-D structural modules exceed this baseline by +22.4%. A future ablation without length_delta would further quantify the length contribution to the structural signal.

5.3 Limitations

Limitation	Impact	Mitigation
Domain specificity (halueval_qa only)	Results may not generalize to dialogue, summarization, or code	Track expansion planned (rationalization_binary, halueval_summarization)
C/B ratio 1.29 (not 1.00)	Residual length signal in lexical baseline	Ablation without length_delta recommended
NIG source-unaware	NIG cannot detect entity-level divergence vs. source	NIG marked DIAGNOSTIC_ONLY; excluded from tribunal vote
744 source pairs	Pilot-scale corpus; low N for statistical power	Results consistent across validation/test; plan for 2,000+ pairs
Lexical baseline not zero	AUC 0.749 remains non-trivial	All reports include baseline side-by-side; delta always reported

5.4 Methodological Boundary

"SAS structural modules (Flow, CRE, Negation) produce reproducible, interpretable structural signal in declarative factual QA. The best composite achieves test $F1 = 0.8717$, precision = 0.9952, recall = 0.7755, and accuracy = 0.8859 on the R0.5D corpus, a +22.4% improvement over the established lexical baseline (AUC 0.749)."

This work does **not** claim:

- Production-grade hallucination detection ready for deployment
- Universal SAS validation beyond halueval_qa declarative QA
- Superiority over all possible lexical or learned baselines
- $kD = 0.56$ as a universal physical or information-theoretic constant
- Scientific validity of runtime-derived features (excluded from all claims)
- Source-aware NIG divergence signal (NIG runs source-unaware here)

6. Conclusion

6.1 Summary

- Structural signal is real and detectable. SAS modules (Flow, CRE) produce reproducible signal exceeding the lexical baseline by +22.4%.
- Flow is the dominant module (F1 = 0.8176, AUC = 0.8457, precision = 1.0000).
- Optimal composite: Flow + CRE + Negation, score ≥ 1 (F1 = 0.8717, precision = 0.9952, recall = 0.7755, accuracy = 0.8859).
- Negation and NIG are domain-specific diagnostics in QA — not independent votes.
- R0-bis cluster analysis holds: correlated modules must be treated as evidence clusters in the tribunal.

6.2 Research Chain Summary

Phase	Core question answered	Key result
R0	Is the infrastructure reproducible?	Yes — F1 $\approx$ 1.0 on 152k pairs
R0-bis	Are modules independent?	No — 3 linearly redundant pairs; cluster design required
R0.5P-1	Can external-clean be generated for prompt-like sources?	Yes — 1,197/1,200 accepted
R0.5P-2A	Can numerical false-presuppositions be paraphrased?	Yes — 1,152/1,200 accepted
R0.5D	Does the declarative corpus reduce the length confound?	Yes — AUC 0.957 $\rightarrow$ 0.749
R1	Do modules produce structural signal?	Yes — F1 = 0.8717 (baseline AUC = 0.9963 dominated)
R1-D	Does structural signal exceed lexical baseline without confound?	Yes — +22.4% over AUC 0.749

6.3 Next Steps

- **R0.5D track expansion** — rationalization_binary or halueval_summarization to test structural signal beyond factual QA
- **Ablation without length_delta** — quantify residual length contribution to structural signal
- **Comprehensive paper** — full SAS standard paper integrating all milestones from R0 through R1-D with formal methodology
- **API and SDK update** — reflect R1-D thresholds and claim boundaries after publication
- **BDI paper (separate)** — Behavioral Divergence Index as SAS extension for temporal model behavior auditing

6.4 Cross-Model Validation

Adversarial auditing of each milestone in the research chain was performed using multiple large language models — including Claude, ChatGPT, DeepSeek, and Gemini — as hostile reviewers. Each auditor independently flagged methodological risks, data leakage patterns, overclaiming, and implementation bugs before results were frozen. Convergence across models is reported as consistency with independent review, not as independent experimental replication. All audit findings and applied patches are documented in the public artifact trail.

6.5 Acknowledgements

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Appendix A: Formal Derivation of $kD = 0.56$

Five independent mathematical convergences support $kD = 0.56$ as a structural criticality threshold. Each path arrives at approximately 0.56 through independent reasoning.

A.1 Statistical Thermodynamics

$$\begin{aligned} T_c &= 2 / \ln(1 + \sqrt{2}) \approx 2.269 \\ kD &\approx 1 - 1/T_c \approx 1 - 0.4407 = 0.5593 \approx 0.56 \end{aligned}$$

A.2 Percolation Theory

$$p_c = 1 / (\text{mean}_k - 1) \approx 1 / 1.8 \approx 0.556 \text{ for } \text{mean}_k \approx 2.8$$

A.3 Rate-Distortion Theory

$$D / D_{\text{crit}} \approx 0.56 \text{ (where } R(D) \rightarrow 0, \text{ lossless limit)}$$

A.4 Lyapunov Stability

$$\lambda_{\text{max}} = 0 \text{ at } \kappa \approx 0.56$$

A.5 Golden Ratio Harmonic

$$kD = (1/\phi) \times (1/\sqrt{2}) \times 1.28 \approx 0.618 \times 0.707 \times 1.28 \approx 0.559 \approx 0.56$$

Appendix B: Corpus Details and Runner Configuration

B.1 R0.5D Corpus Statistics

Attribute	Value
Track	halueval_qa (factual QA)
B filter	B_hallucination >= 15 words
Original corpus	10,000 pairs
After filter	2,510 eligible pairs
Selected for generation	1,200
Generations OK	1,197 (99.75%)
C_clean accepted	744 (62%)
Total paired rows (x2)	1,488
C/B length ratio (mean)	1.29
C/B length ratio (median)	1.24
C/B token Jaccard (mean)	0.36
Lexical baseline AUC (test)	0.749
Generator version	r05d_generator.py v0.1.5
Model	anthropic/claude-3-haiku via OpenRouter
Verifier version	r05d_verifier.py v0.1.4
long_b_ngram_overlap threshold	4 (adjusted from 2 — topic overlap in QA)

B.2 R1-D Runner Configuration

Parameter	Value
Script	r1_module_runner.py v1.0.7
TDA mode	full
MSC	skip-msc (DIAGNOSTIC_ONLY)
NIG status	DIAGNOSTIC_ONLY_STANDALONE (source-unaware)
Total rows processed	1,488
Runtime (approx.)	14.45 seconds
Batch size	500
Errors	0

B.3 Artifact Integrity

Artifact	SHA-256
r1_eval_archive.zip	41B6BF24440E04E579DA57298783BDDFF5294CA73D10BBB14516DF34E135A699
R1D_COMPLETE_20260709.zip	24E69B7D20F190389ABAC1737268C54ADA23EBFD4CCF54E4A8888AAFBF944C90

Artifact	SHA-256
r1d_results_archive.zip	37F2B540737C11B18622A586611D2DA90057722678148691AC8E77F9538E6191
R0.5D companion ZIP	67171A10DD3E56791DF3F2EE8F23F09C241F3EFBA93EBCD6F82863A46BBC171C